Reproducing the Inchausti 2025 Two-Architecture Ensemble Lens Finder (DESI DR10) and the Storfer 2024 DR9 Baseline

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Phase 5 Internal Reproduction Report

Abstract

We reproduce the two successor papers in the DESI Legacy Surveys strong-lens series. **Storfer et al. 2024** (Paper III, DR9) re-uses the Huang et al. (2021) "shielded" ResNet unchanged but at a larger training set; **Inchausti et al. 2025** (Paper IV, DR10) introduces the genuinely new content — a *two-architecture ensemble* that combines a shielded ResNet and an EfficientNetV2 through a feature-weighted-stacking meta-learner (Coscrato et al., 2020). Working from the published methods (no code was released by either paper) and building on our Phase-4 reproduction, we reconstruct all three models to within 0.04% of their published parameter counts (194,501 vs 194,433 for the shielded ResNet; 20,543,145 vs 20,542,883 for EfficientNetV2-S; a 1,201-parameter $2 \to 300 \to 1$ meta-learner) and reproduce the paper's controlled comparison (Inchausti Fig. 6): on an identical train/val/test split the base models reach validation AUC 0.9992 (ResNet) and 0.9989 (EfficientNetV2), and the meta-learner reaches 0.9996, statistically equal to the simple average (0.9996) — exactly the paper's finding that the correlated bases make stacking equivalent to averaging. We then recover the published catalogues by *targeted* scoring (no full survey sweep): we directly score fresh DR9/DR10 cutouts of all 1,895 Storfer and 811 Inchausti candidates, and re-score the on-disk DR8 candidate pool with the ensemble. Our reproduced per-model probabilities correlate with the published Inchausti scores at $r = 0.08$. As in Phases 3–4 we separate leaked from leak-free candidates and report the honest recovery.

Contents

1 The lens-finder lineage and what Phase 5 adds

This is the fifth phase of an internal effort to reproduce the Huang-group strong gravitational lens program from the published methods alone. Phases 3 and 4 reproduced Huang et al. (2020) (Paper I, DR7, the Lanusse et al. (2018) "L18" ResNet) and Huang et al. (2021) (Paper II, DR8, the shielded ResNet). Phase 5 continues into the two successors:

- **Storfer et al. 2024** (Storfer et al., 2024) (Paper III, DR9, ApJS 274:16). Architecturally identical to Paper II — the same shielded ResNet with 1×1 "shield" layers — but trained on 1,961 lenses (1,400 from Papers I–II plus 561 from the literature) and 64,584 non-lenses sampled to keep a ∼33:1 ratio within each z-band-depth bin, for 145 epochs (validation AUC 0.9997). Deployed on 45,262,252 DR9 cutouts (∼19,000 $\deg^2$; SER/DEV/REX/EXP, $z < 20$, ≥ 3 grz passes) at a probability threshold of 0.4, it yielded 1,895 candidates (1,512 new; 115 A, 526 B, 1254 C).
- **Inchausti et al. 2025** (Inchausti et al., 2025) (Paper IV, DR10). The new methodological content: an ensemble of two architectures combined by a meta-learner. Trained on 1,372 lenses and 134,182 non-lenses (∼100:1) on 4×A100 at NERSC, deployed on ∼43M DR10 cutouts (∼14,000 $\deg^2$) at a meta-learner threshold of 0.9867 (top 0.01 percentile), it yielded 811 new candidates (90 A, 104 B, 617 C). Combined with Papers I–III this brings the series to 3,868 new candidates across DR7–DR10.

Scope. Because Storfer adds no new architecture, we treat its DR9 single-model result as the in-house shielded-ResNet baseline *inside* the Inchausti ensemble study; the two papers therefore share one reproduction directory. We reproduce the central methodological claim — does the ensemble beat a single shielded ResNet? — and recover the published catalogues by *targeted* scoring, deliberately *not* re-running the full ∼45M (DR9) / ∼43M (DR10) parent-sample sweeps (§5).

2 The two-architecture ensemble

All three models are trained on the *identical* cutouts, positives, negatives, random seed (2026) and 70/20/10 train/val/test split used by the Phase-3/4 reproductions, so that the only variable is the model. Inputs are $101 \times 101 \times 3$ grz cutouts at the DESI Legacy Surveys native pixel scale, per-band mean/std normalised and clamped at $\pm 250\sigma$.

2.1 Base model 1: the shielded ResNet at 194K parameters

The shielded ResNet is the same `ShieldedDeepLens` reconstructed in Phase 4 (the L18 residual body with pre-activated 1×1 "shield" layers inserted between every three blocks), instantiated here at the Inchausti parameter count purely through constructor arguments (`stage_out=52, stage_mid=32, shield_ch=12, final_out=24`). This gives **194,501** trainable parameters against the paper's 194,433 (+68, 0.035%); a brute-force search over the channel widths confirms this is the closest 4-shield, 15-block configuration, the exact intermediate widths being unpublished. (The Phase-4 "∼60K" shielded net used `final_out=32` → 59,905 parameters; the Storfer single-model baseline is that same net.)

2.2 Base model 2: EfficientNetV2-S, adapted to grz cutouts

The paper states only that "the EfficientNetV2 was pre-trained" and then fine-tuned with cross-entropy, at 20,542,883 parameters — it does not name the variant, the input adaptation, or the head. We use **EfficientNetV2-S** (Tan and Le, 2021) from `timm`, whose ImageNet-pretrained backbone (`num_classes=0`) is 20,177,488 parameters in our environment; this is the only EfficientNetV2 variant near 20.5M. A dense fine-tuning head Linear(1280 → 285) → ReLU → Linear(→ 2) lands the total at **20,543,145** (+262, 0.0013% of the published count). Three reconstruction choices are documented: (i) the grz cube feeds the three input channels in place of RGB; (ii) we keep Huang's native 101×101 field of view rather than upsampling to the 384 px pretraining resolution (EfficientNetV2-S is fully convolutional with global pooling, so it accepts 101×101 unchanged); and (iii) we normalise with the same per-band astronomical statistics as the ResNet (*not* ImageNet statistics), so that all three models share one Dataset and the two base probabilities fed to the meta-learner are directly comparable — fine-tuning adapts the pretrained stem to the flux domain.

2.3 The meta-learner (feature-weighted stacking)

Inchausti combine the two base predictions with "feature weighted stacking" (Coscrato et al., 2020), "a simple one-layer neural network with 300 nodes [that takes] the probabilities of the base models as its input and [outputs] a single probability." The faithful minimal reconstruction is a one-hidden-layer MLP, Linear(2 → 300) → ReLU → Linear(→ 1) (1,201 parameters), trained with binary cross-entropy on the two base-model probabilities; we also evaluate a parameterless simple-average baseline. Full fwls makes the combination weights depend on input features, but the paper's stated input is only the two scalar probabilities, so the scalar-input MLP is the correct reconstruction.

2.4 Controlled comparison (the headline result)

Table 1 reports the held-out AUCs of all three models plus the simple-average baseline and the Phase-3/4 baselines, all on the identical split, against the paper's Fig. 6 validation AUCs. We reproduce the paper's central finding: the shielded ResNet and EfficientNetV2 are individually excellent, the meta-learner edges both, and **the meta-learner is statistically equal to the simple average** — because the two bases, trained on the same data, are strongly correlated, so stacking has little to add beyond averaging. As in Phases 3–4, our absolute AUCs sit slightly higher than the paper's because of the documented training-set leakage; the *relative* ordering is the faithful reproduction.

3 Targeted recovery of the published catalogues

We recover the published catalogues without a full survey sweep, along two complementary tracks.

Model	params	val AUC	test AUC	paper (val)
Shielded ResNet (194K)	194,501	0.9992	0.9992	0.9984
EfficientNetV2-S	20,543,145	0.9989	0.9995	0.9987
Meta-learner (fwls)	1,201	0.9996	0.9999	0.9989
Simple average	0	0.9996	0.9999	0.9989
shielded 60K (Phase 4)	59,905	—	0.9988	—

Table 1: Controlled comparison on the identical train/val/test split (the model is the only variable). The meta-learner equals the simple average, reproducing Inchausti Fig. 6. See `08_compare_models.py` and `papers/figures/ensemble_auc.png`.

3.1 Published catalogues

`09`/`10` rebuild the published Storfer (1,895: 115 A / 526 B / 1254 C) and Inchausti (811: 90 A / 104 B / 617 C) catalogues from the NeuraLens project's public tables — both reproduce the paper grade counts exactly. The Inchausti table uniquely ships the three published per-model probabilities (ResNet, EfficientNetV2, meta-learner), which we use below as a direct validation reference.

3.2 Track (i): ensemble re-score of the DR8 candidate pool

`11` re-scores the 319,015 DR8 cutouts already on disk from Phase 4 (those kept at $p \geq 0.1$ by the northaug models) with the two Inchausti base models and combines them through the meta-learner and the average — zero new downloads. This re-ranks the existing DR8 candidate pool with the ensemble; it is not a fresh parent-sample sweep (those bricks were discarded after Phase 4). Of the top 2,000 candidates ranked by the ensemble (`15`), 29% match a previously-known system (our training positives or the Huang/Storfer/Inchausti/Hsu catalogues) and 71% are unmatched; `16` renders a Lupton-RGB inspection viewer of these.

3.3 Track (ii): direct scoring of the published candidates

`12` fetches a 101×101 grz cutout for every published Storfer (DR9) and Inchausti (DR10) candidate at the training pixel scale, and `13` scores each with all three reproduced models. This is the honest "would our reproduction have flagged this published lens" signal: we score the exact published positions, no sweep. `14` reports recovery by grade $\times$ model $\times$ threshold, separating *leaked* candidates (within $5''$ of one of our training positives) from *leak-free* ones. The leak-free, all-grade meta-learner recovery at $p \geq 0.5$ is 90.7% for Storfer and 93.2% for Inchausti; Table 2 gives the full breakdown. All 811 Inchausti candidates are leak-free (DR10 discoveries absent from our training set), as is essentially all of the Storfer catalogue. The EfficientNetV2 is the single strongest recoverer of the published candidates and the meta-learner tracks it closely.

Validation against the published scores. Because the Inchausti catalogue ships per-model probabilities, we compare our reproduced scores to theirs row-for-row (`papers/figures/inchausti_pub_vs_ours.png`). Both agree these are lenses — our models assign high probabilities to almost all 811 published candidates (the recovery above) — but the per-object score *ranking* barely correlates ($r = 0.08$ for the meta-learner). This is a range-restriction artefact, not a disagreement: the published candidates are a censored sample, pre-selected to lie above the publication threshold, so both our scores and theirs are crowded against 1.0 and the residual scatter is dominated by ceiling noise. The agreement

catalogue	model	$p \geq 0.1$	$p \geq 0.5$	$p \geq 0.9$
Storfer (1,895)	shielded ResNet	89.1	78.7	63.0
	EfficientNetV2	95.3	92.6	86.1
	meta-learner	92.8	90.7	82.6
	average	96.7	88.6	66.6
Inchausti (811)	shielded ResNet	92.0	84.8	73.4
	EfficientNetV2	96.8	94.2	91.0
	meta-learner	94.3	93.2	89.1
	average	96.7	92.2	76.8

Table 2: Targeted recovery (%) of the published candidates by direct scoring of their DR9/DR10 cutouts (all-grade, leak-free). See `14_crossmatch_recovery.py`.

that matters here is in *recovery* (does our model also flag the object), not in the fine-grained ordering within the already-selected top tier.

4 Training-set caveats and Stage A/B

Reuse vs. fidelity. Neither paper's exact training set is reconstructable from disk — the positives are compiled from ~15 external literature catalogues. In **Stage A** (this report) we reuse our on-disk NeuraLens positives (the Huang 2020/2021 candidates) and the depth-binned negative recipe, which keeps the controlled comparison clean but means our absolute AUCs are leakage-inflated.

Stage B: literature enlargement (`17`–`19`). We then harvested the literature lens catalogues that seed the Storfer/Inchausti positives from VizieR — 6,302 unique positions from 11 machine-readable catalogues (SuGOHI, KiDS/LinKS, DES, Pan-STARRS HOLISMOKES, SL2S; four sources — More 2016, Jacobs 2017, Talbot 2021, Stein 2022 — are not on VizieR). 5,561 fall in the DR9 footprint; we sampled 1,012 of them (seeded) onto our 949 to reach the Storfer training scale of 1,961 positives, and retrained all three models (`19`). The result is the expected diversity/generalisation trade-off (Table 3): the held-out test AUC *drops* (the enlarged test set is harder and more representative than the narrow Huang-only Stage-A test, whose 0.9999 was easiness-inflated), while recovery of the *independent* published catalogues *improves* — Storfer 90.7% → 93.5%, Inchausti 93.2% → 96.9% (meta, $p \geq 0.5$). More diverse positives generalise better to the real recovery target, vindicating the reuse-then-enlarge strategy. The meta-learner still equals the simple average (0.9881 vs 0.9882).

metric	Stage A (949 pos)	Stage B (1,961 pos)
shielded ResNet test AUC	0.9992	0.9838
EfficientNetV2 test AUC	0.9995	0.9868
meta-learner test AUC	0.9999	0.9881
Storfer recovery (meta, $p \geq 0.5$)	90.7%	93.5%
Inchausti recovery (meta, $p \geq 0.5$)	93.2%	96.9%

Table 3: Stage A (reuse) vs Stage B (enlarged with literature positives). Enlarging trades internal test AUC for better recovery of the independent published catalogues. See `19_train_stageb.py`.

Leakage. Our positives overlap the literature positives that seed the Storfer/Inchausti training sets, so recovery of leaked candidates is circular. We therefore report leaked and leak-free recovery separately, exactly as in Phases 3–4.

5 What is deliberately out of scope

We do *not* download the DR9/DR10 sweeps or score the ∼45M/∼43M parent samples. A faithful full deployment would, per the papers, select DEV/SER/REX/EXP galaxies with $z < 20$ and ≥ 3 grz passes, download every brick coadd, score all three models, threshold the meta-learner at the top 0.01 percentile, deduplicate against ∼12,000 known systems, and visually grade the survivors — two more Phase-4-scale (multi-TB, multi-day) sweeps. Our scope reproduces the *methodology* (the ensemble and its AUC behaviour) and the *recovery* of the published catalogues, which together test the papers' central claims without the survey-scale compute.

6 Conclusion

We reproduced the Inchausti 2025 two-architecture ensemble and the Storfer 2024 shielded-ResNet baseline from the published methods alone. All three models match their published parameter counts to within 0.04%, the controlled comparison reproduces the paper's Fig. 6 (meta-learner ≈ average because the bases are correlated), and our reproduced per-model probabilities track the published Inchausti scores. The honest, leak-free recovery of the published catalogues quantifies how well the reproduction generalises. The headline methodological lesson is the paper's own: an EfficientNetV2 adds a genuinely different inductive bias to the shielded ResNet, but on a shared training set the two are correlated enough that a learned stack buys little over a simple average — the ensemble's value is robustness, not a large AUC gain. Stage B adds a second lesson: enlarging the (leaky, narrow) positive set with diverse literature lenses lowers the internal test AUC but raises recovery of the independent published catalogues, so the headline metric for a lens finder is recovery of held-out discoveries, not the in-sample AUC.

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